Financing Frictions: Theory

One wedge, six primitives

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Outline

General Setting

Micro-Foundations of the Wedge

One period, one input: a friction makes the firm invest too little

Static economy, one period, a representative firm

Technology in capital alone (labor held fixed):

$$y_i = z_i \, k_i^{\alpha}, \qquad \alpha \in (0, 1)$$

- Why do we care?

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- Frictionless economy

  $\rightarrow$ All NPV>0 projects are financed

  $\rightarrow$ Who is rich and who is productive does not matter

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  $\rightarrow$ All NPV> 0 projects are financed

  $\rightarrow$ Who is rich and who is productive does not matter

$\rightarrow$ This lecture: friction makes firm i choose k_i below level a frictionless firm with same z_i

Frictionless benchmark: equal returns, net worth irrelevant

Entrepreneur with net worth a_i funds the shortfall $k_i - a_i$ at the common rate r:

$$\max_{k_i \geq 0} \underbrace{z_i k_i^{\alpha}}_{\text{revenue}} - \underbrace{r\,(k_i - a_i)}_{\text{external finance}} - \underbrace{r\,a_i}_{\text{own funds}} = z_i k_i^{\alpha} - r k_i$$

First-order condition and optimal scale:

$$\underbrace{\alpha z_i k_i^{\alpha-1}}_{\text{MRPK}_i} = r \qquad \Longrightarrow \qquad k_i^* = \left(\frac{\alpha z_i}{r}\right)^{\frac{1}{1-\alpha}}$$

Net worth a_i cancels: every firm reaches k_i^*, so $\text{MRPK}_i = r$ for all i (Modigliani and Miller, 1958)

One friction, one wedge: $\text{MRPK}_i = r + \lambda_i$

Every core friction reopens the same gap in the benchmark relation:

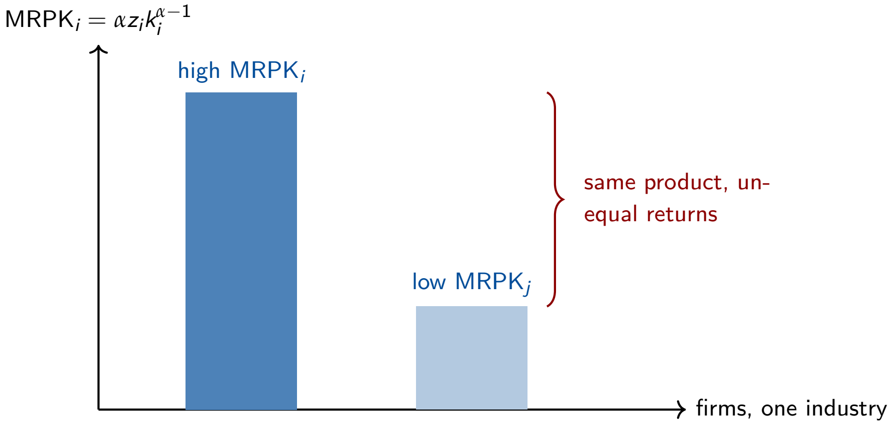
$$\text{MRPK}_i = r + \lambda_i, \qquad \lambda_i \geq 0$$

$\lambda_i =$ shadow wedge that keeps firm i below its frictionless scale

- $\lambda_i > 0 \Rightarrow$ profitable marginal investment firm cannot fund at cost r

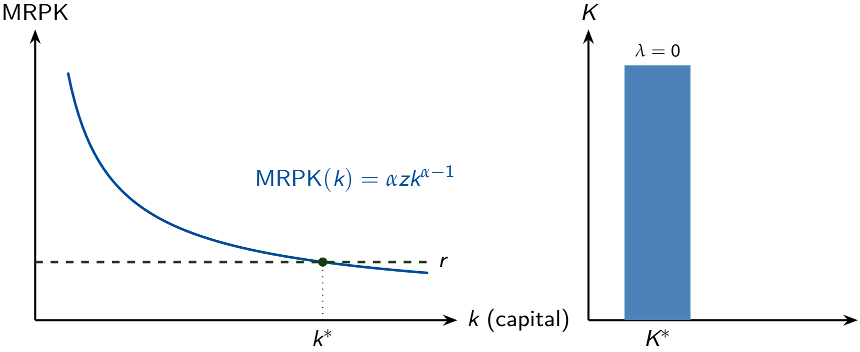
$\rightarrow$ Which benchmark assumptions must fail for $\lambda_i > 0$?

Capital earns very different returns despite similar productivity



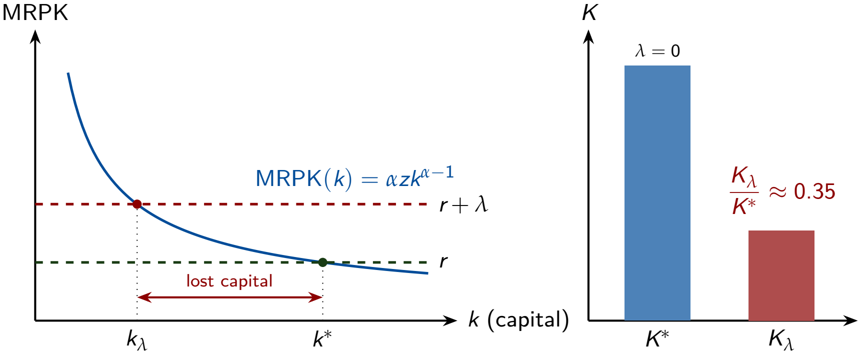
Why we care: a positive wedge sharply shrinks the capital stock

Frictionless firm invests to MRPK $= r$



Why we care: a positive wedge sharply shrinks the capital stock

Frictionless firm invests to MRPK $= r$ Wedge firm stops at MRPK $= r + \lambda$



$\frac{K_\lambda}{K^*} = \left(\frac{r}{r+\lambda}\right)^{\frac{1}{1-\alpha}}$: doubling the cost of capital ($\lambda = r$, $\alpha = \frac{1}{3}$) leaves 35% of the stock cost

The wedge takes two forms: quantity vs price

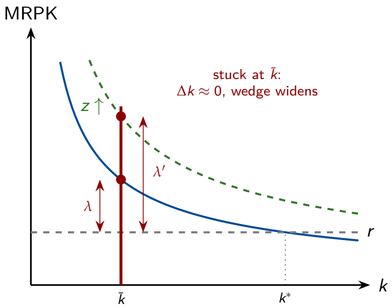
- Quantity rationing: firm held at a corner, off its first-order condition
- Price wedge: firm stays interior, paying a higher marginal cost of funds

Takeaway 1

One underinvestment wedge, two forms: a quantity ceiling (firm at a corner) or a price premium (firm interior)

Same underinvestment, opposite reaction to a productivity shock

Quantity rationing



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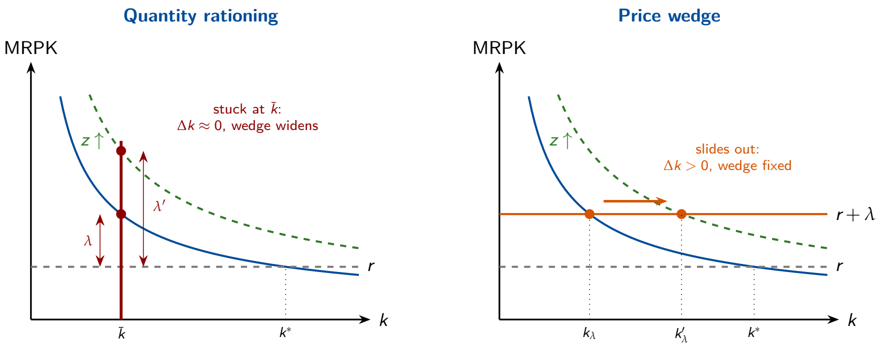

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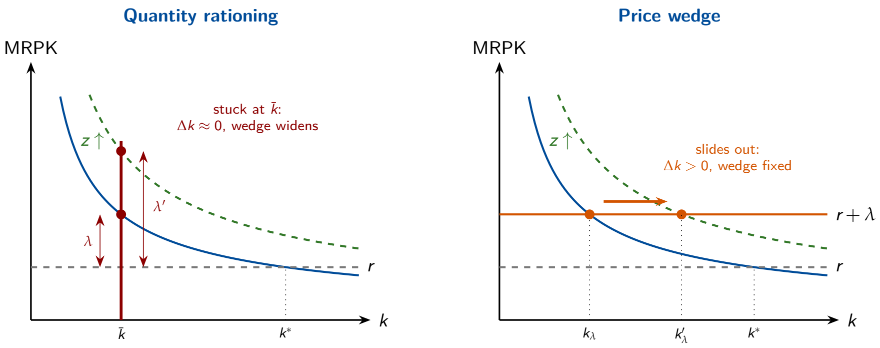

Takeaway 2

Same underinvestment, opposite response to a shock: holding net worth fixed, whether investment tracks productivity identifies the regime in the data

The two regimes hide in different data and call for different levers

Quantity rationing: wedge = shadow cost never paid in the loan rate

→ Two firms at the same rate can face different λ_i

Price wedge: firm actually pays $r + \lambda_i$, so the wedge shows up directly in the spread; identified by comparing exposed and unexposed borrowers

Takeaway 3

Read the wedge off the right data: MRPK_i dispersion under rationing (never the loan rate), the spread under a price wedge

Most credit is collateralized, and collateral does not track productivity

$\text{Loan} = f(\text{cash flow}, \text{collateral})$

- cash-flow-based lending is forward-looking; asset-based lending is backward-looking (Brooks and Dovis, 2020)

- most credit is collateralized: Europe 70% (Degryse et al., 2025); Spain 40%, Peru 43% (Ivashina et al., 2022)

Takeaway 4

Collateral, not productivity, gates credit (Finlay, 2025): nothing makes the productive firm the one with collateral, so rationing tracks the productivity-wealth gap and misallocation arises

We derive limited enforcement next, the most transparent of the six generators of a collateral ceiling

Outline

General Setting

Micro-Foundations of the Wedge

Primitives open the wedge; ceilings and premia are their shadows

Primitive frictions: deep information, enforcement, or market-structure problems that block borrowing at the first-best rate and scale

Reduced-form borrowing technologies: the tractable objects those primitives generate, collateral ceilings, leverage caps, external finance premia

Six primitives, one wedge

#	Mechanism	What breaks	Side	Anchor model
1	Limited enforcement / collateral	Commitment	Borrower	Kiyotaki and Moore (1997)
2	Moral hazard / pledgeability	Hidden action	Borrower	Holmstrom and Tirole (1997)
3	Adverse selection / rationing	Hidden type	Borrower	Stiglitz and Weiss (1981)
4	Costly state verification	Hidden output	Borrower	Townsend (1979)
5	Intermediary constraints	Scarce bank capital	Lender	Gertler and Kiyotaki (2010)
6	Lender market power / hold-up	Imperfect competition	Lender	Rajan (1992)

Today we derive row 1 end to end; the other five share the wedge but break a different assumption

One model in full

Limited enforcement and collateral

Friction 1 breaks enforcement: the borrower can default and walk away

The benchmark assumed perfect enforcement; drop it, so courts enforce repayment only up to seizable collateral

Date-zero balance sheet, capital installed from own funds plus a loan:

$$k_i = a_i + b_i$$

If the entrepreneur defaults, the lender recovers only a fraction $\theta \in [0, 1]$ of installed capital:

$$\text{recovery on default} = \theta\, k_i$$

This θ is the deep primitive behind the reduced-form ceiling $b_i \leq \theta k_i$

Date 1: the loan is settled in cash at the gross rate $(1 + r)\, b_i$

Repayment incentive compatibility caps borrowing at θk_i

Repaying and continuing must beat defaulting and walking away:

$$\underbrace{z_i k_i^\alpha + k_i - (1+r) b_i}_{\text{repay, keep capital}} \geq \underbrace{z_i k_i^\alpha + (1-\theta) k_i}_{\text{default, abscond with } (1-\theta) k_i}$$

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Revenue $z_i k_i^{\alpha}$ cancels on both sides; the capital terms net to θk_i:

$$(1+r)\, b_i \leq \theta k_i$$

The ceiling grows with net worth and pledgeability

Substitute $b_i = k_i - a_i$ into $(1+r)b_i \leq \theta k_i$:

$$(1+r)(k_i - a_i) \leq \theta k_i \qquad \Longrightarrow \qquad k_i \leq \bar{k}_i^{LE} \equiv \frac{(1+r)\,a_i}{(1+r)-\theta}$$

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- $\theta = 0$: pure self-finance, $\bar{k}_i^{LE} = a_i$

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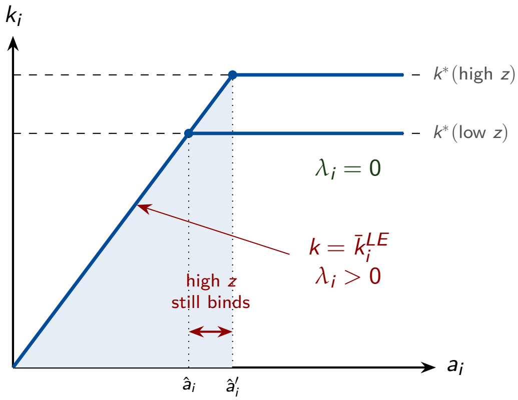
- higher θ or higher a_i relaxes the ceiling ✓ higher r tightens it ✗
- $\theta = 0$: pure self-finance, $\bar{k}_i^{LE} = a_i$
- $\theta = 1$: full pledgeability, $\bar{k}_i^{LE} = \frac{(1+r)a_i}{r}$, finite, still not k^*
  - $\rightarrow$ Low-a_i firm still constrained despite perfectly pledgeability $\rightarrow$ why?

Constraint is the gap between the frictionless size and the observed size



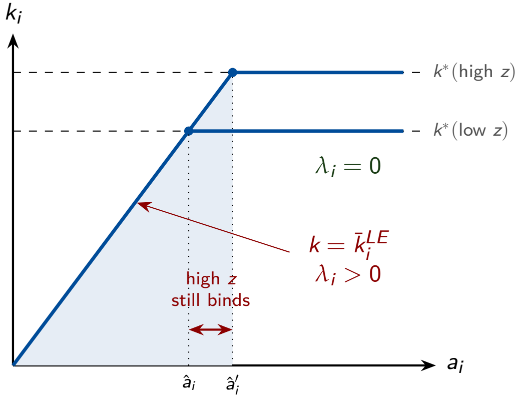
Firm invests $k_i = \min\{k_i^*, \, \bar{k}_i^{LE}\}$

Binds whenever net worth is too small:

$$a_i < \frac{(1+r)-\theta}{1+r} \, k_i^*$$

High z_i, low a_i, low θ push a firm into the constrained region

Constraint is the gap between the frictionless size and the observed size



Key takeaway

Low net worth firms can be unconstrained if their productivity is small

High net worth firms can be constrained if their productivity is very high

The shadow wedge λ_i^{LE} rises as net worth falls

For a constrained firm, plug $k_i = \bar{k}_i^{LE}$ into $\text{MRPK}_i = \alpha z_i k_i^{\alpha-1}$:

$$\text{MRPK}_i = \alpha z_i \left(\frac{(1+r) - \theta}{(1+r)\, a_i} \right)^{1-\alpha} > r$$

so the wedge is

$$\lambda_i^{LE} = \alpha z_i \left(\frac{(1+r) - \theta}{(1+r)\, a_i} \right)^{1-\alpha} - r$$

Larger when z_i is high, a_i is low, θ is low, the same directions that govern the ceiling

Constrained firms are young and asset-light

Most exposed: firms with low net worth relative to efficient scale, and firms whose assets are hard to pledge

- intangible-intensive and software firms
- human-capital-intensive and services firms
- young firms with little retained wealth

Least exposed: asset-heavy firms with high liquidation value, at the same productivity and net worth

Levers raise pledgeable value or net worth

Because the ceiling is $\bar{k}_i^{LE} = \frac{(1+r)a_i}{(1+r)-\theta}$, policy works through θ or a_i:

- stronger creditor rights and contract enforcement, raising θ
- collateral and movable-asset registries, making more assets seizable
- public guarantees that raise effective collateral value
- wealth transfers or start-up grants, raising a_i for high-ability, low-wealth entrants

Friction 3: where Stiglitz-Weiss sits in the map

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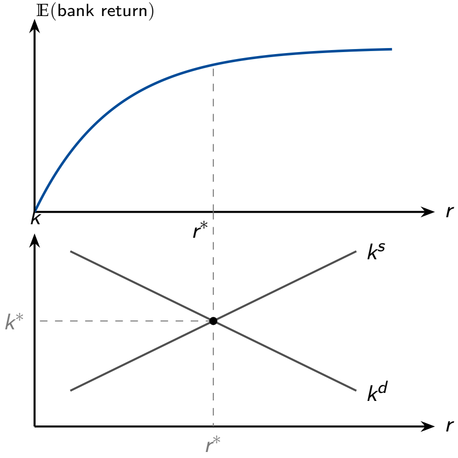
Row 1 (limited enforcement) is derived in full; row 3 gives the sharpest rationing picture, on one slide next

Stiglitz-Weiss in one slide: key intuition

Frictionless: $r^* = r^{\text{walras}}$



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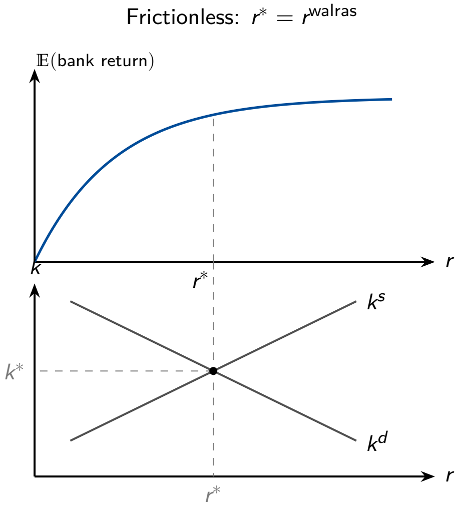

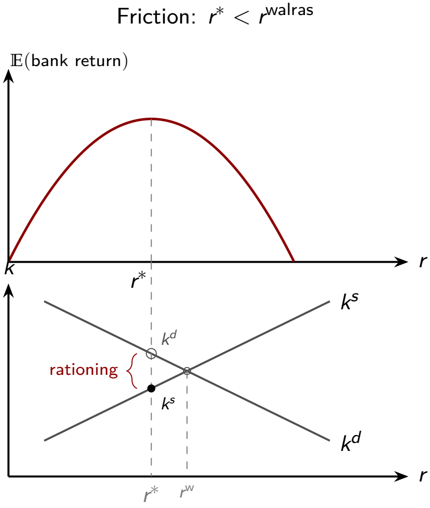

Same wedge, a different lever for each primitive

The wedge λ_i is the same; the right policy lever depends on which primitive opened it:

- Limited enforcement $\rightarrow$ collateral registries, stronger creditor rights
- Adverse selection $\rightarrow$ information sharing, credit bureaus
- Market power $\rightarrow$ bank competition

Key challenge

The same λ_i is consistent with several primitives, so identifying the exact friction in the data is the hard part

Four takeaways from the models

Takeaway 1

One wedge, two forms: a quantity ceiling (corner) or a price premium (interior)

Takeaway 2

Different data reveal each regime: MRPK_i dispersion under rationing, the spread under a price wedge

Takeaway 3

Collateral, not productivity, gates credit, so rationing maps to the productivity-wealth gap and misallocation

Takeaway 4

Same underinvestment, opposite reaction to a shock: rationed firms cannot act on good news, priced firms can

A static wedge is necessary, not sufficient, for misallocation

- benchmark: $\text{MRPK}_i = r$, net worth irrelevant
- friction: $\text{MRPK}_i = r + \lambda_i$, one reduced form, six primitives
- collateral: $\bar{k}_i^{LE} = \frac{(1+r)a_i}{(1+r)-\theta}$, wedge λ_i^{LE} rising as a_i falls
- quantity rationing strands firms at a corner; a price wedge keeps them interior

A static $\lambda_i > 0$ distorts investment today, but self-financing can undo it: persistent misallocation needs forces that keep recreating high-λ_i firms (Moll, 2014)

Thank you!

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